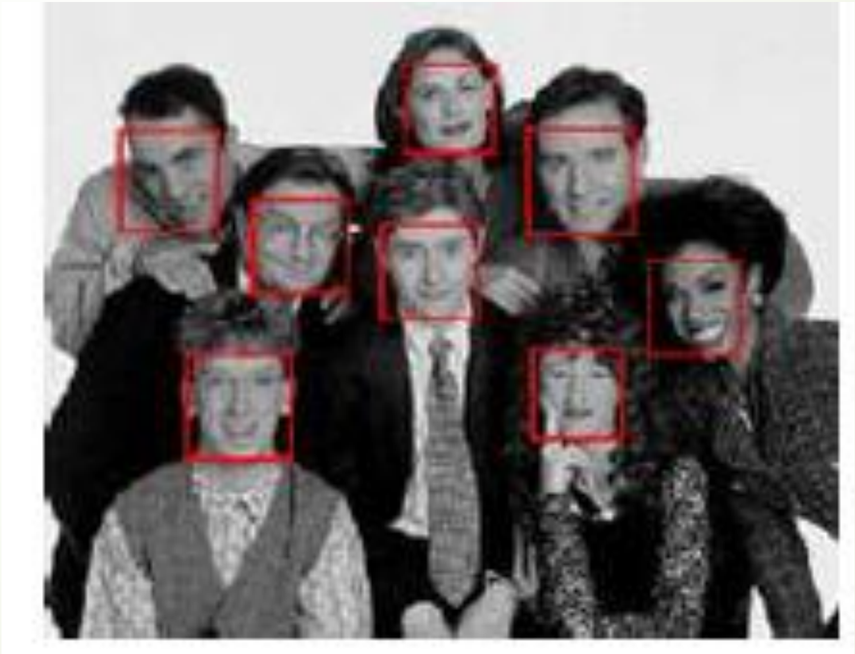
Recognition

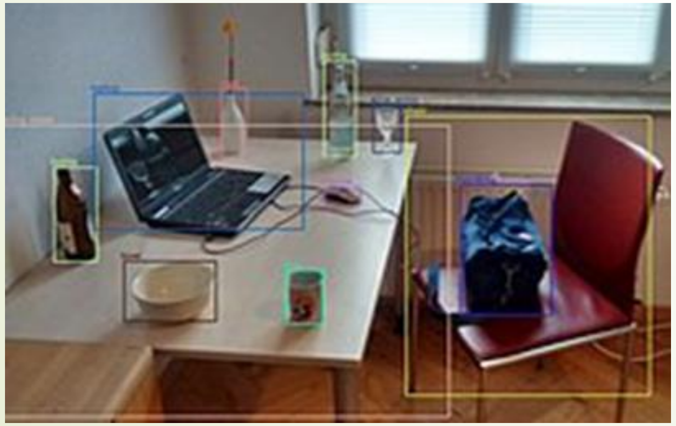
Chapter 7

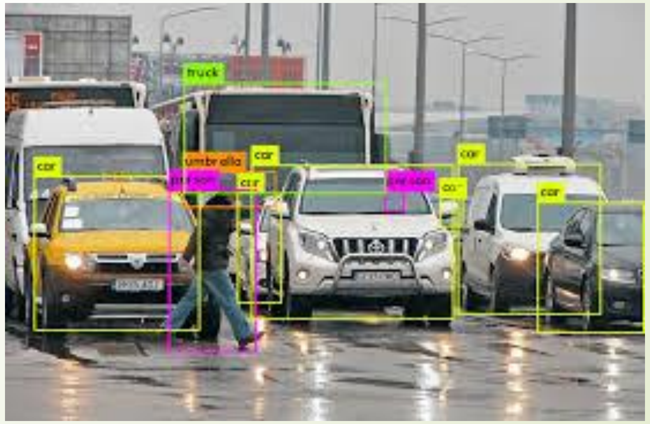


Contents

Object Detection

- Object detection is a computer technology related to computer vision and image processing that deals with detecting instances of semantic objects of a certain class (humans, buildings, or cars) in digital images and videos
- Well-researched domains of object detection include face detection and pedestrian detection.
- Object detection has applications including image retrieval, video surveillance, activity recognition, face detection, face recognition, etc.
- It is also used in tracking objects.





wikipedia.org and researchgate.net

Object Detection

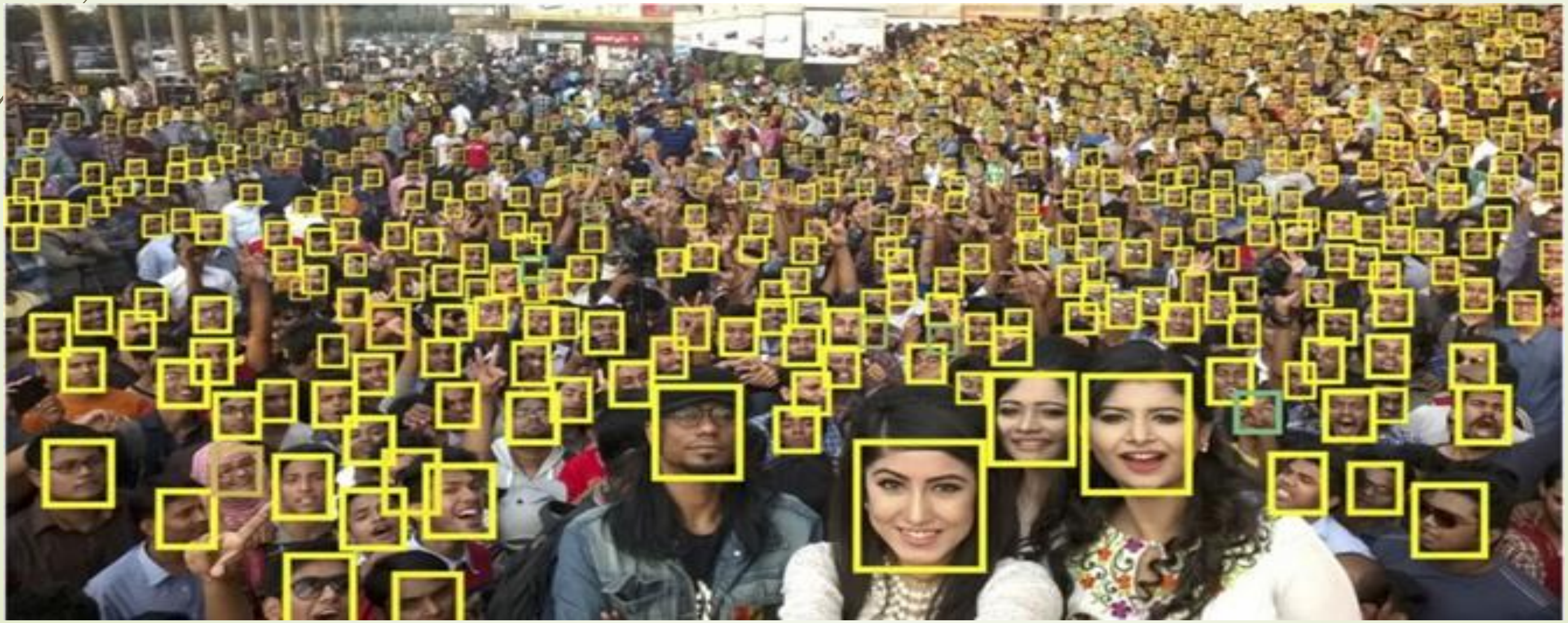
How object detection works?

- The general working of object detection is:

1. Input Image
2. Pre-processing
3. Feature Extraction
4. Classification
5. Localization
6. Non-max Suppression
7. Output

Face Detection

- Face detection refers to the process of automatically locating and identifying human faces within digital images or video frames.
- The goal of face detection is to determine whether or not there are any faces in the image.
    - If present, return the image location and extent of each face.
- Before face recognition can be applied to a general image, the locations and sizes of any faces must first be found.



becominghuman.ai

Face Detection

Importance of face detection

- The first step of face recognition system.
- First step in many Human Computer Interaction systems
  - Expression Recognition
  - Cognitive State/Emotional State Recognition
- First step in many surveillance systems
- A step towards Automatic Target Recognition(ATR) or generic object recognition

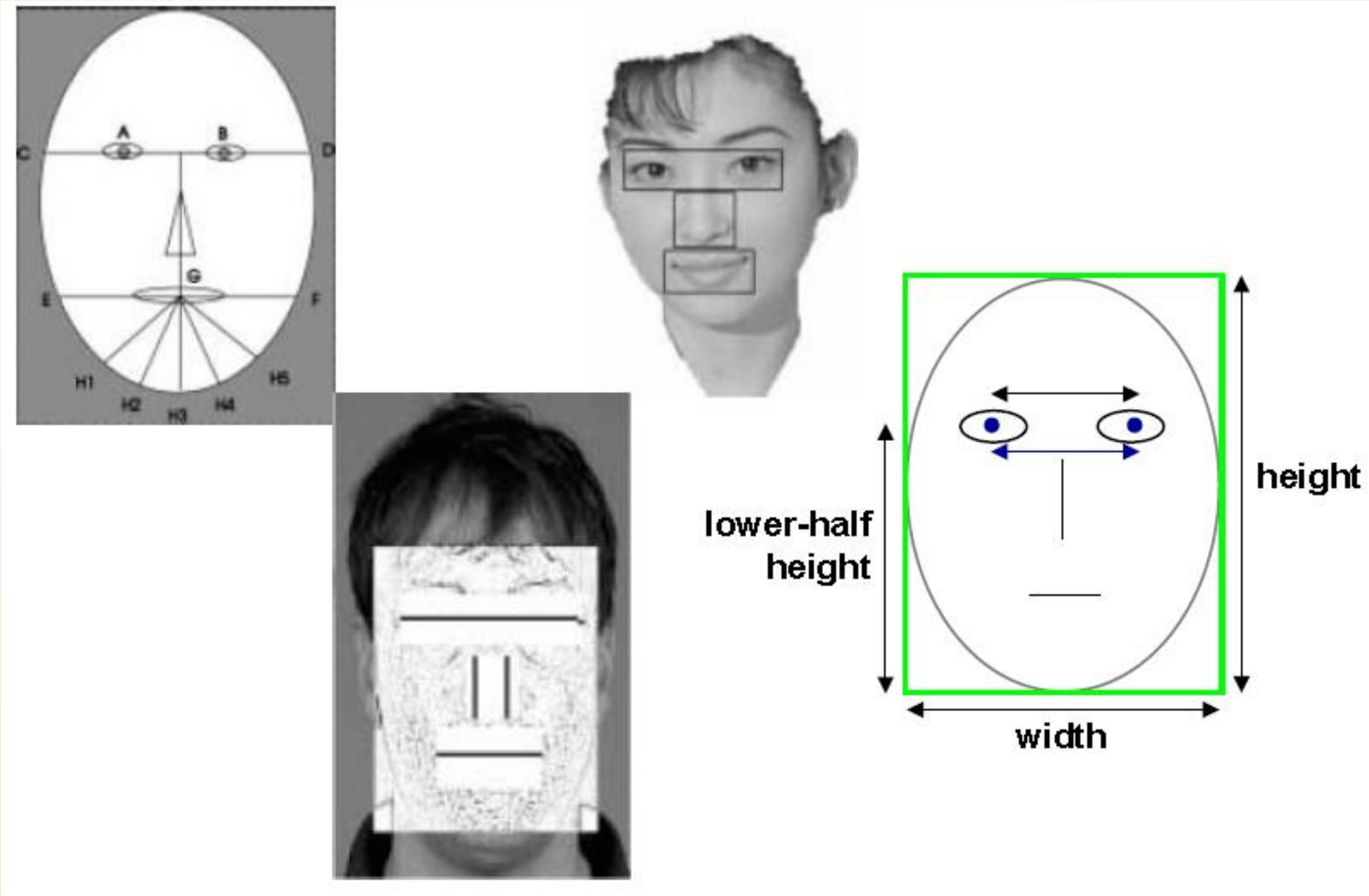
Face detection techniques

- Feature-based
- Template-based
- Appearance-based

Face Detection

Feature-based detection

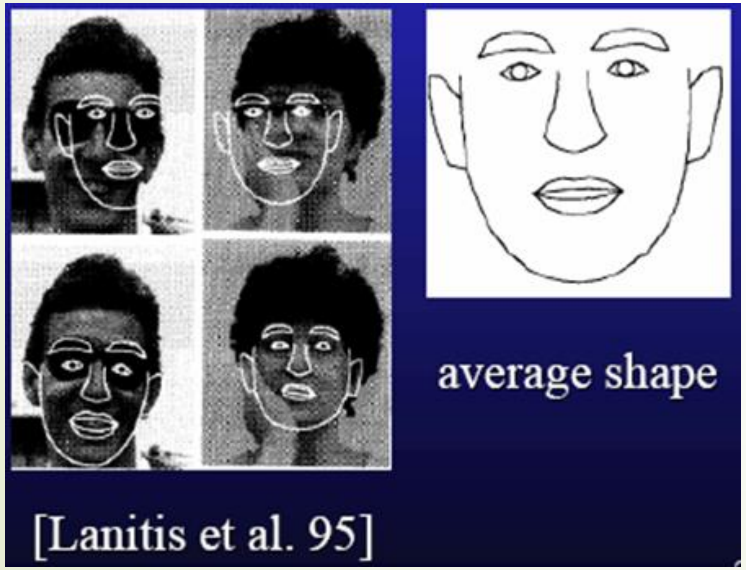
- This methods involve detecting key facial features like eyes, nose, and mouth, and then using their arrangement to identify a face.

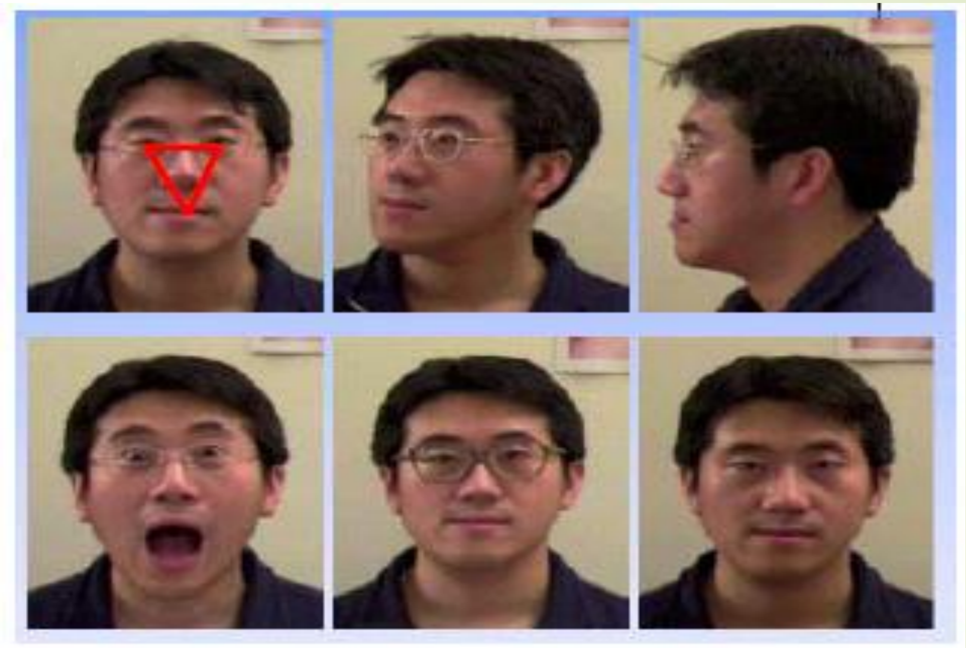


Face Detection

Template-based detection

- It involves comparing a template image of a face with different regions of an input image to find matching regions.
- Template-based approaches can deal with a wide range of pose and expression variability.
- Several standard patterns stored to describe the face as a whole or the facial features separately.





Face Detection

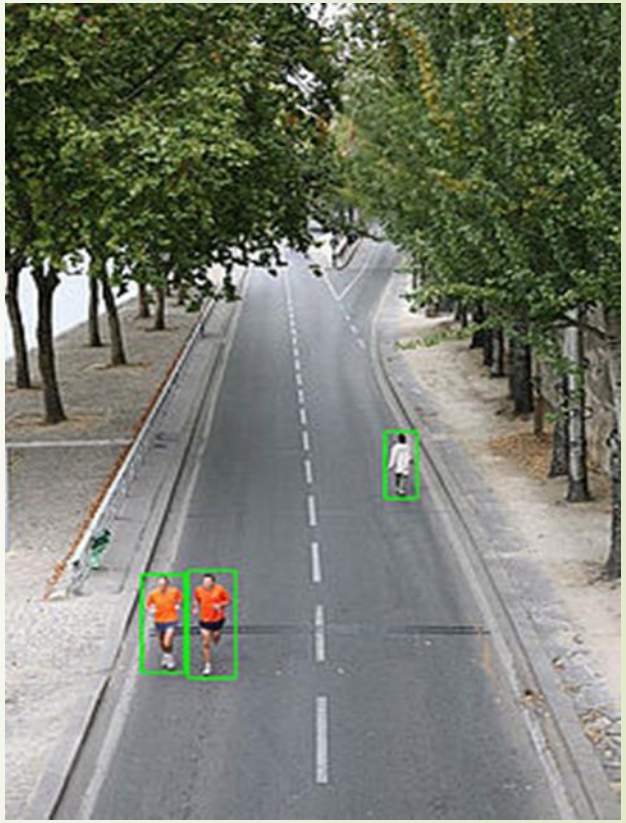
Appearance-based detection

- The models (or templates) are learned from a set of training images which capture the representative variability of facial appearance.
- Most appearance-based approaches today rely heavily on training classifiers using sets of labeled face and non-face patches.
  - Clustering and PCA
  - Neural networks
  - Support vector machines
  - Boosting

Face Detection

Pedestrain detection

- Pedestrian detection is a safety system that uses sensors, cameras, and AI to identify people near vehicles.
- It can alert drivers or automatically apply the brakes
- It is an essential and significant task in any intelligent video surveillance system.
- It has an obvious extension to automotive applications due to the potential for improving safety systems.



Face Detection

Challenges

- Various style of clothing in appearance
- Different possible articulations
- The presence of occluding accessories
- Frequent occlusion between pedestrians

Existing Approaches

- Holistic detection
- Part-based detection
- Patch-based detection
- Motion-based detection
- Detection using multiple cameras

Face Recognition

Difference between Face Detection and Recognition

- Detection – two-class classification
  - Face vs. Non-face
- Recognition – multi-class classification
  - One person vs. all the others

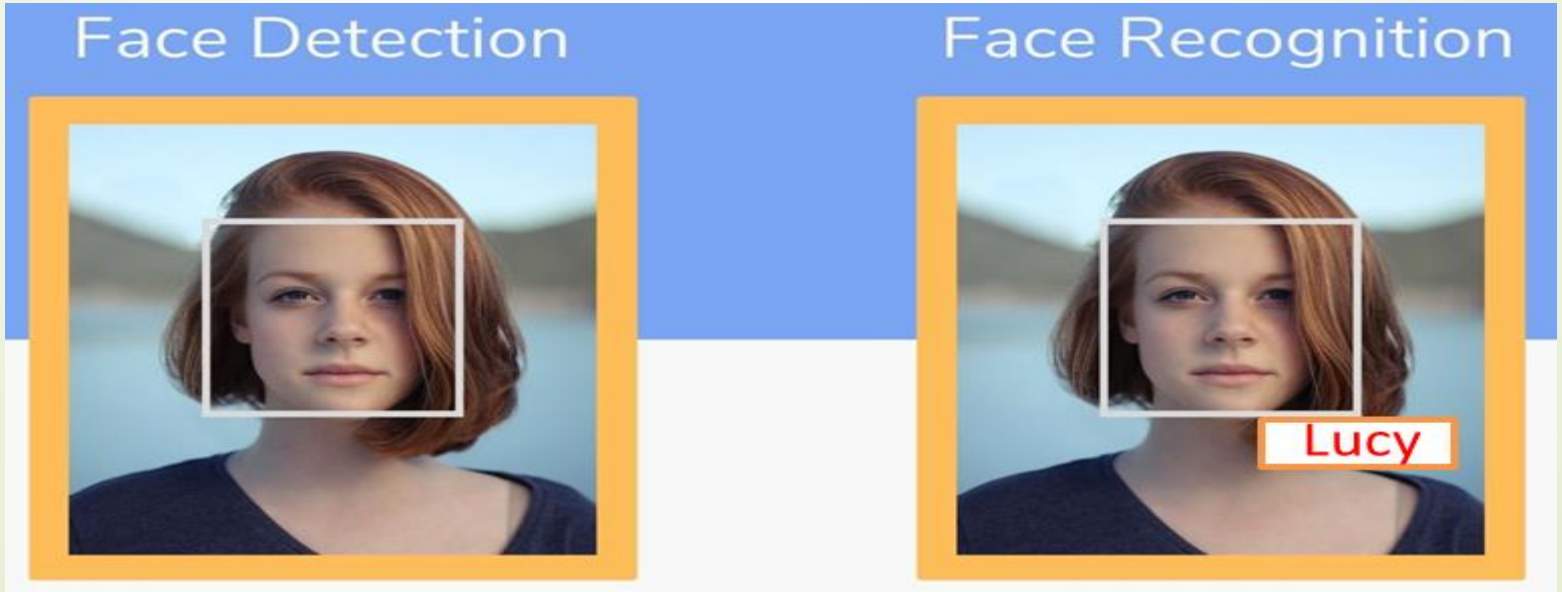


Photo by Christopher Campbell on Unsplash

Face Recognition

What are Biometrics?

- A biometric is a unique, measurable characteristic of a human being that can be used to automatically recognize an individual or verify an individual‟ s identity.
  - Finger-scan
  - Facial Recognition
  - Iris-scan
  - Retina-scan
  - Hand-scan

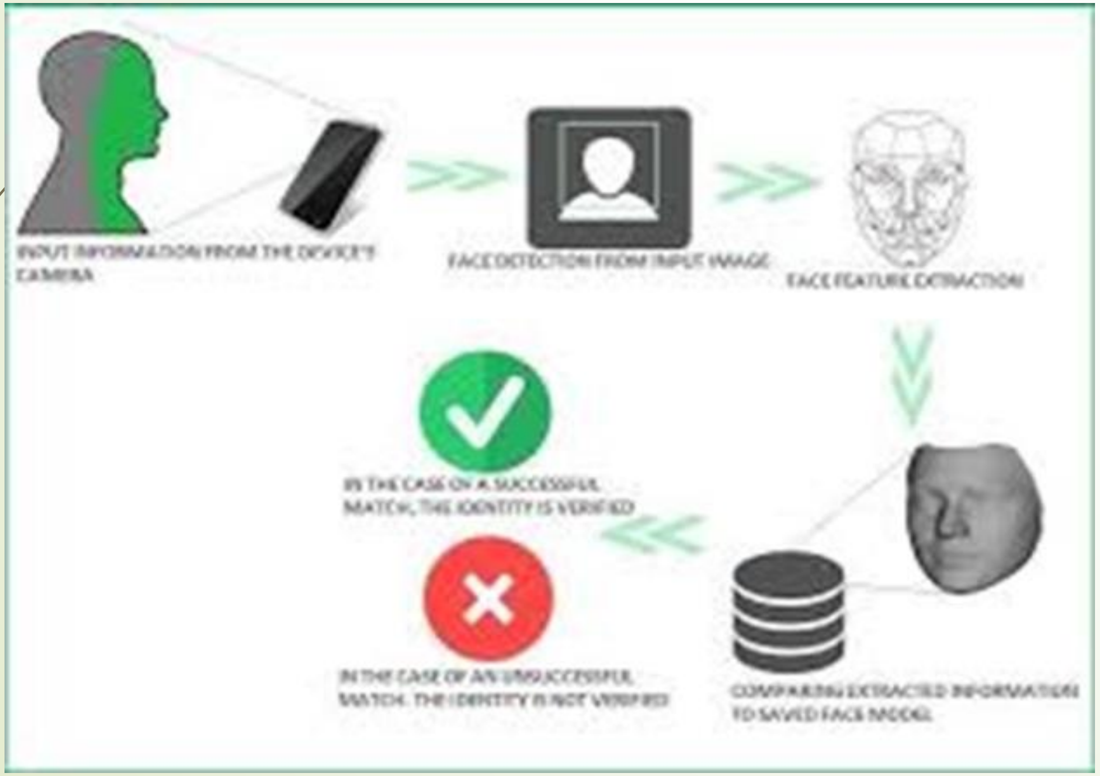
Face Recognition

Why we choose Face Recognition?

- There are number reasons to choose face recognition. This includes the following :
  - It requires no physical interaction on behalf of the user.
  - It is accurate and allows for high enrolment and verification rates.
  - It does not require an expert to interpret the comparison result.
  - It can use your existing hardware infrastructure
  - It is the only biometric that allow you to perform passive identification in a one to.

Face Recognition

- Face recognition is a method of identifying or verifying the identity of an individual using their face.
- Face recognition systems can be used to identify people in photos, video, or in real-time.



- The key idea is to compare the face of an image to images in a database of stored records and establish whose fact it is.

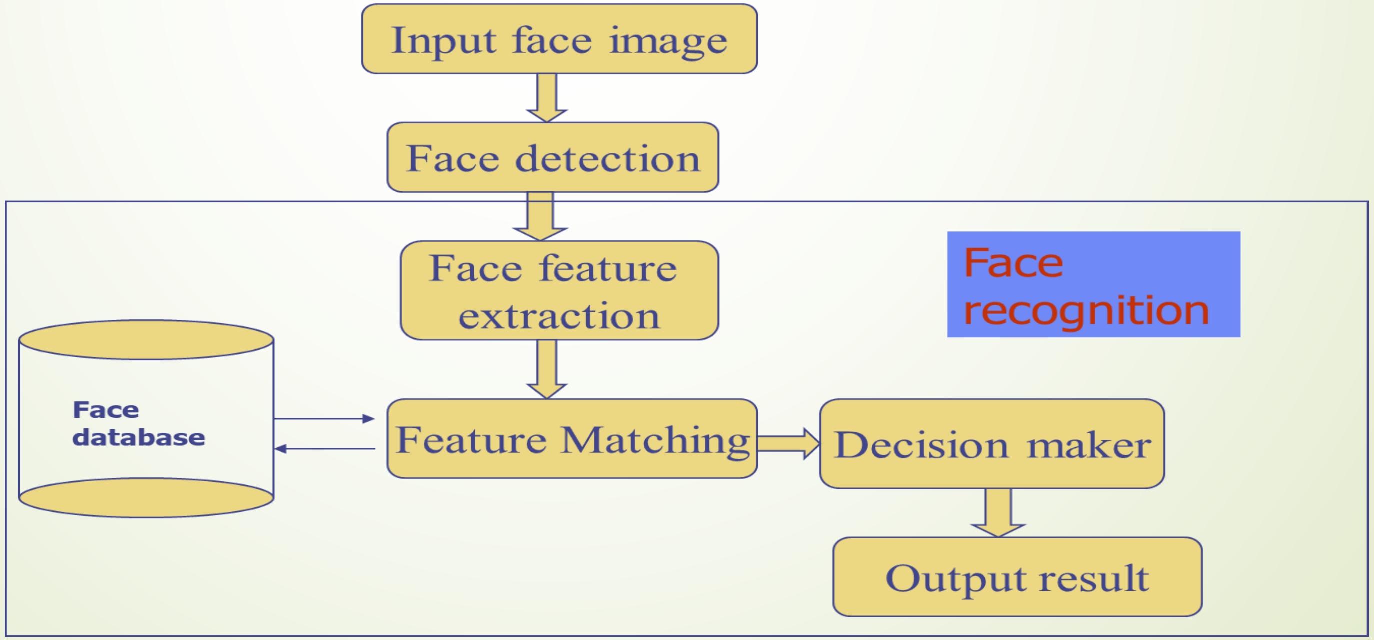
Face Recognition

- It can operate in either or both of two modes:
- *Face verification* (or authentication): involves a one-to-one match
  - that compares a query face image against a template face image whose identity is being claimed.
- *Face identification* (or recognition): involves one-to-many matches
  - that compares a query face image against all the template images in the database to determine the identity of the query face.

Face Recognition

Basic step of face recognition

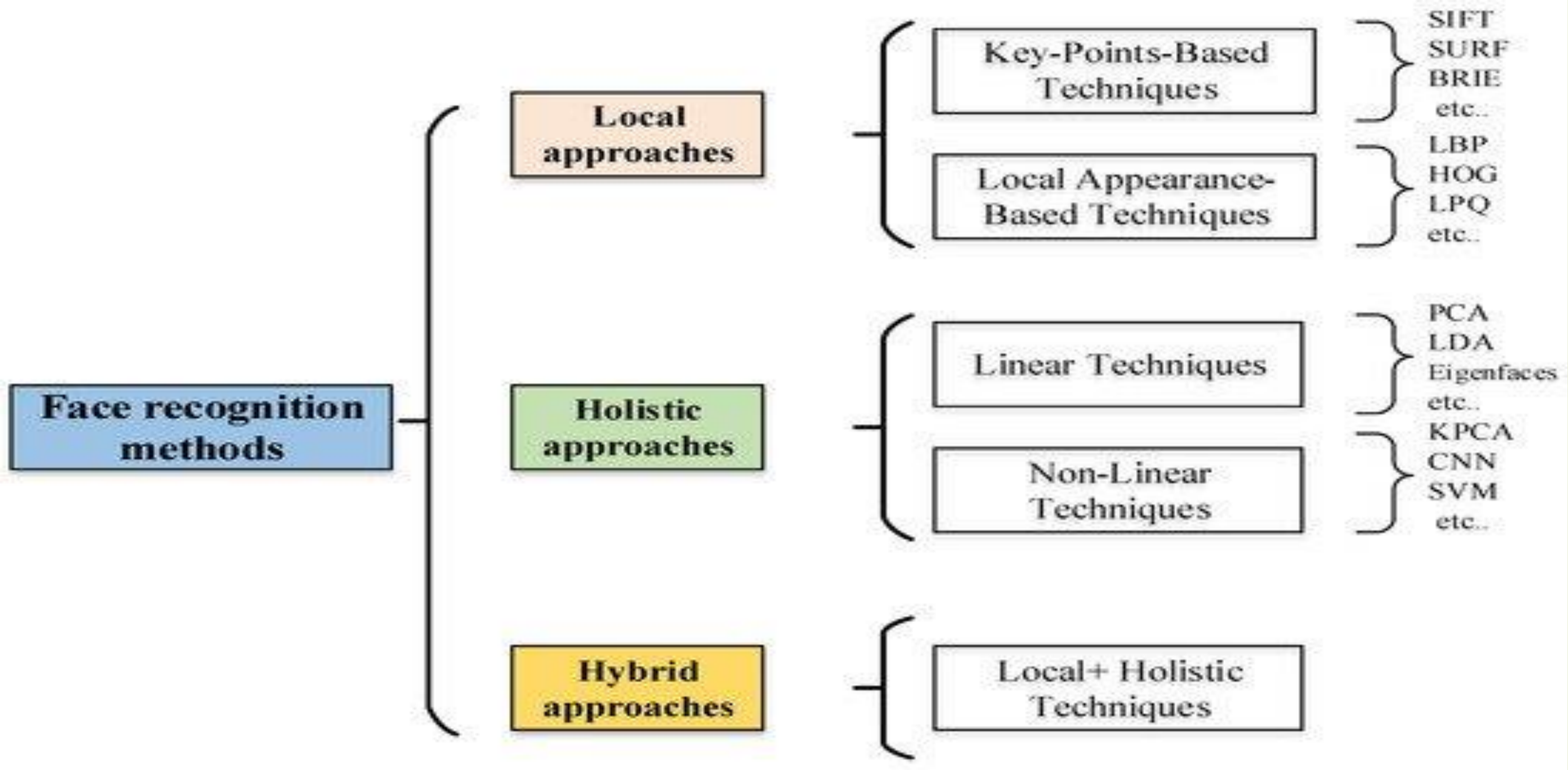
- Face recognition has many potential applications.
- For many years not very successful, it is needed to improve the accuracy of face recognition
- Combining face recognition and other biometric recognition technologies, such as:
  - fingerprint recognition technology,
  - voice recognition technologies and so on



Face Recognition

Approaches

- Numerous face recognition methods/algorithms have been proposed in last 20 years,



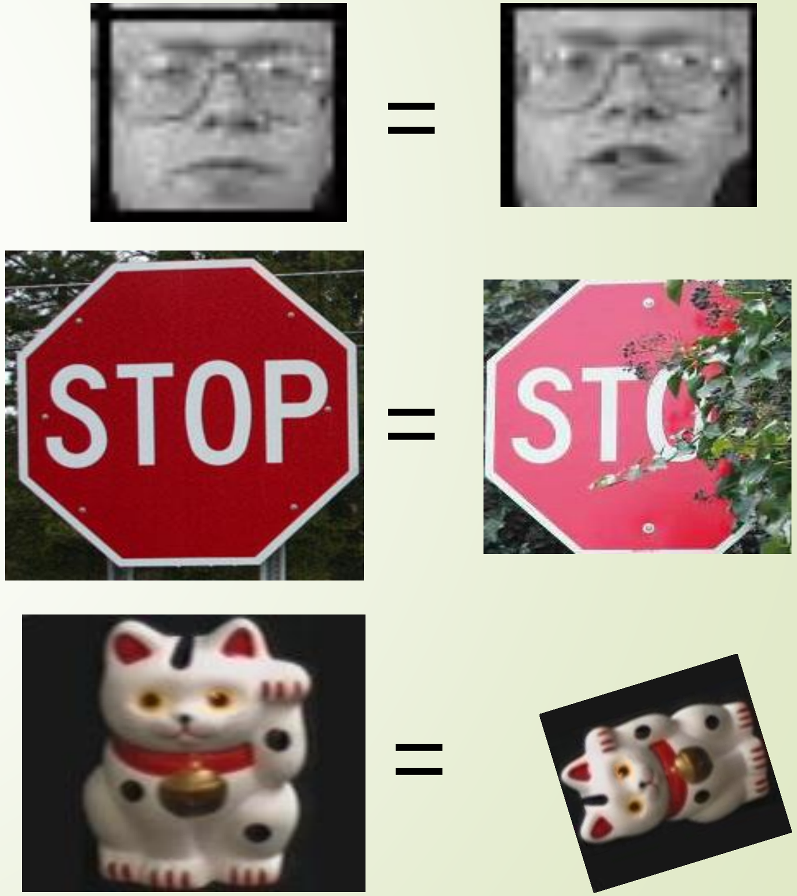
Face Recognition

Applications

- The natural use of face recognition technology is the replacement of PIN.
- Government Use:
  - Law Enforcement
  - Security/Counterterrorism
  - Immigration
  - Voter verification
- Commercial Use:
  - Residential Security
  - Banking using ATM
  - Physical access control of buildings areas, doors, cars or net access

Object Instace Recognition

- Want to recognize the same or equivalent object instance, which may vary
  - Slight deformations
  - Change in lighting
  - Occlusion
  - Rotation, rescaling, translation, perspective



Object Instace Recognition

Geometric Alignment

- To recognize one or more instances of some known objects,
- first extracts a set of interest points in each database image and stores the associated descriptors.
- At recognition time, features are extracted from the new image and compared against the stored object features.
- Whenever a sufficient number of matching features are found for a given object,
  - the system then invokes a match verification stage,
  - to determine whether the spatial arrangement of matching features is consistent with those in the database image.

Object Instace Recognition

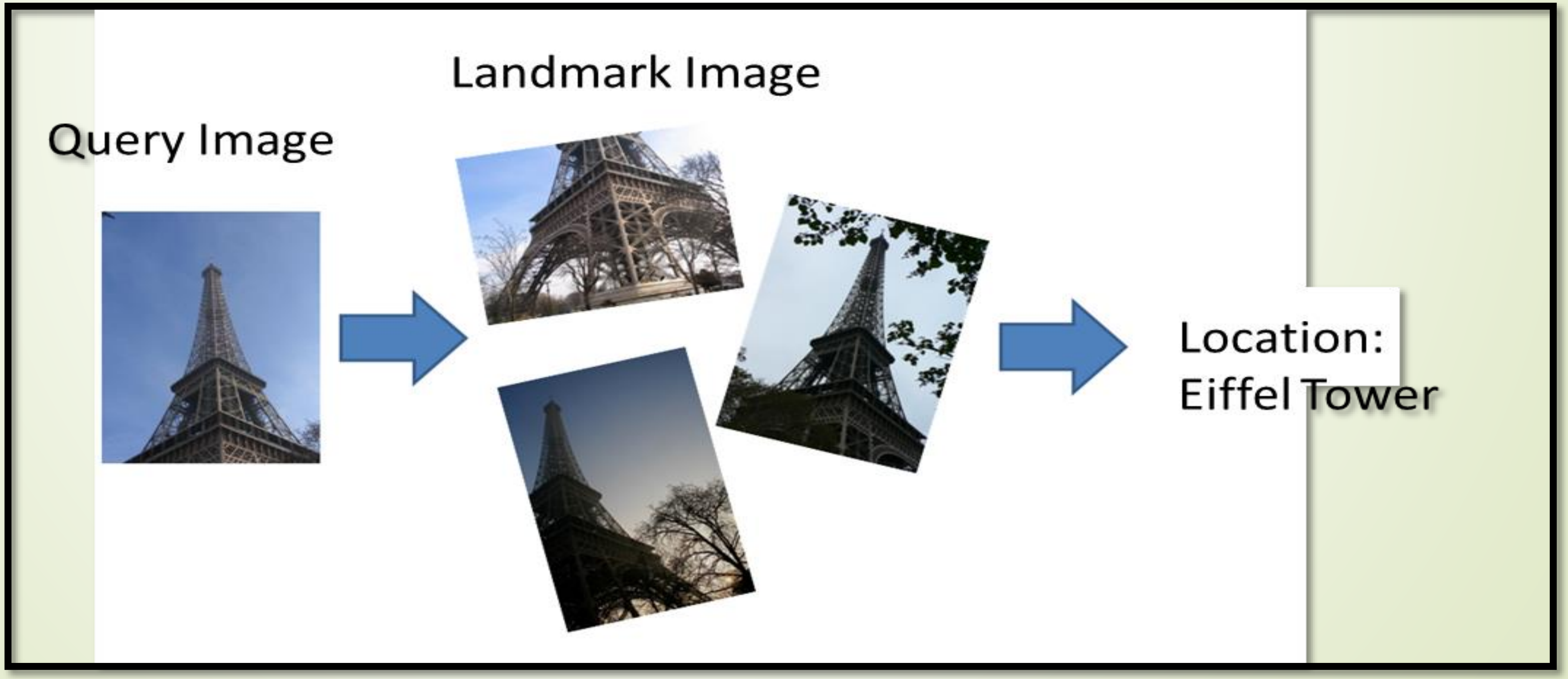
Geometric Alignment

- Recognizing objects in a cluttered scene (Lowe 2004) c 2004 Springer.
  - Two of the training images in the database are shown on the left.
- They are matched to the cluttered scene in the middle using SIFT features, shown as small squares in the right image.

Object Instace Recognition

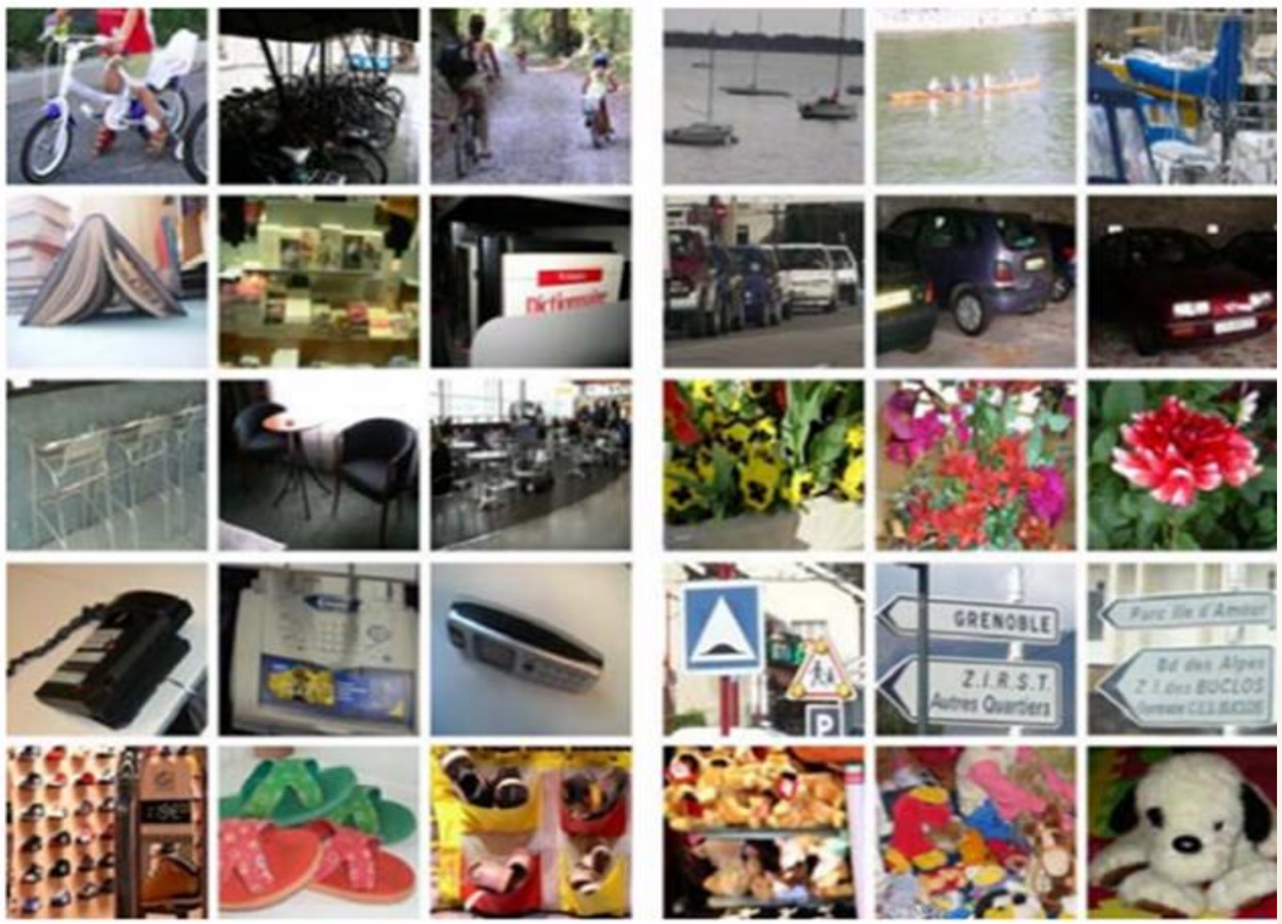
Application: Location Recognition

- One of the most exciting applications of instance recognition today is in the area of location recognition
  - Used both in desktop applications and in mobile applications



Category Recognition

- Category recognition, also known as *object classification*, is the process of identifying and organizing objects into categories based on their visual features
- Consider for example the set of photographs:10 different visual categories



How would you go about writing a program to categorize each of these images into the appropriate class?

Visual category recognition is an extremely challenging problem.

Category Recognition

How it works?

- *Image classification*: Labels images with one or more object types
- *Image annotation*: Labels images with all objects present
- *Object detection*: Finds the location of objects in images
- *Image parsing*: Identifies every object and surface class in an image
- *Image understanding*: Describes the content of an image

Techniques

- *Histogram of Oriented Gradients (HOG):* Extracts gradient orientation histograms from images
- *Color Histograms*: Analyzes the color distribution in images
- *Convolutional Neural Networks (CNNs):* Use layers of convolutional filters to automatically extract features and classify objects
- *Transfer Learning*: Uses pre-trained CNN models

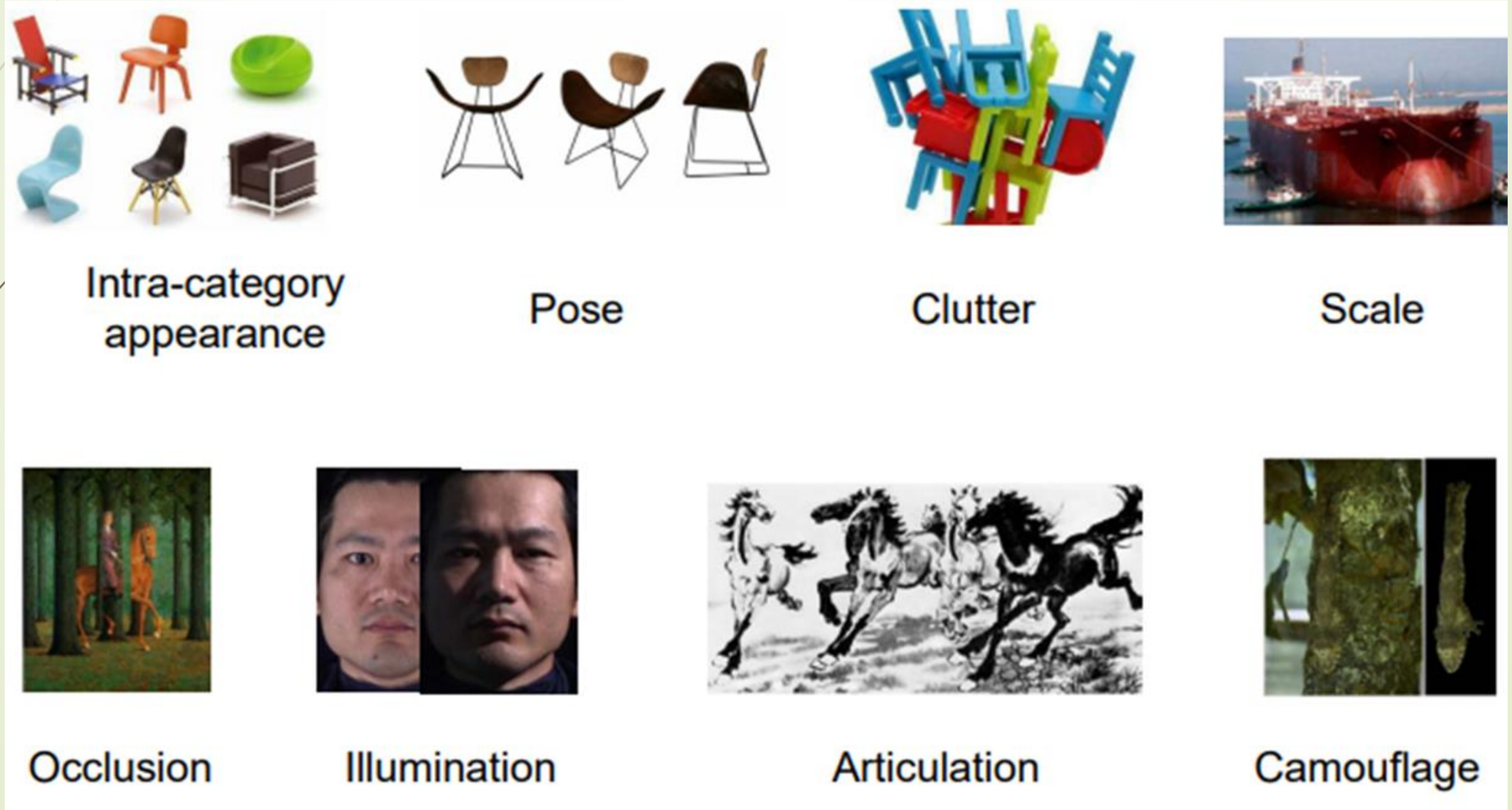
Category Recognition

Challenges

- *Visual variability*: Objects within the same category can appear very different due to factors like lighting, pose, and deformation.
- *Occlusion*: When parts of an object are hidden by other objects, it becomes difficult for the model to identify the category correctly.
- *Viewpoint changes*: Different viewing angles of the same object can significantly alter its appearance, posing a challenge for recognition.
- *Scale variations*: Objects appearing at different sizes in an image can confuse the model.
- *Background clutter*: Distinguishing the target object from a complex background with many irrelevant details can be challenging.
- *Limited training data*: Sometimes, it might not be enough data available for certain categories.
- *Class imbalance*: Some categories have significantly more training data than others.
- *Complexity of human recognition*: Humans can readily recognize a vast number of object categories, which is challenging to replicate with computer vision models.

Category Recognition

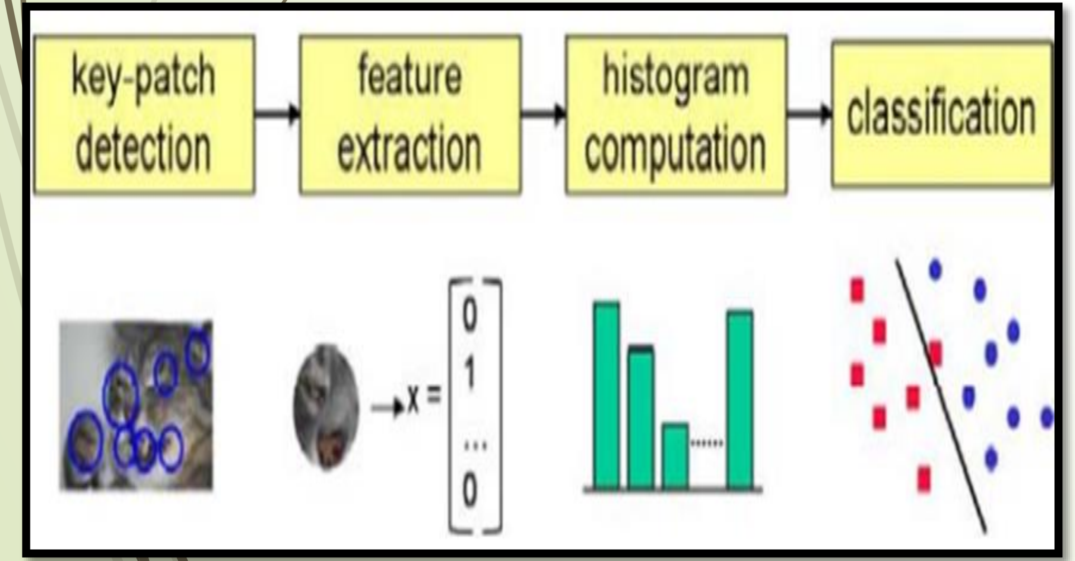
Challenges



Category Recognition

Bag –of-Feature Approach

- One of the simplest algorithms for category recognition is the bag of words (also known as bag of features or bag of keypoints)
- That represent objects and images as unordered collections of feature descriptors.
- Computes the distribution (histogram) of visual words found in the query image and compares this distribution to those found in the training images.
- Excellent results in the presence of background clutter.

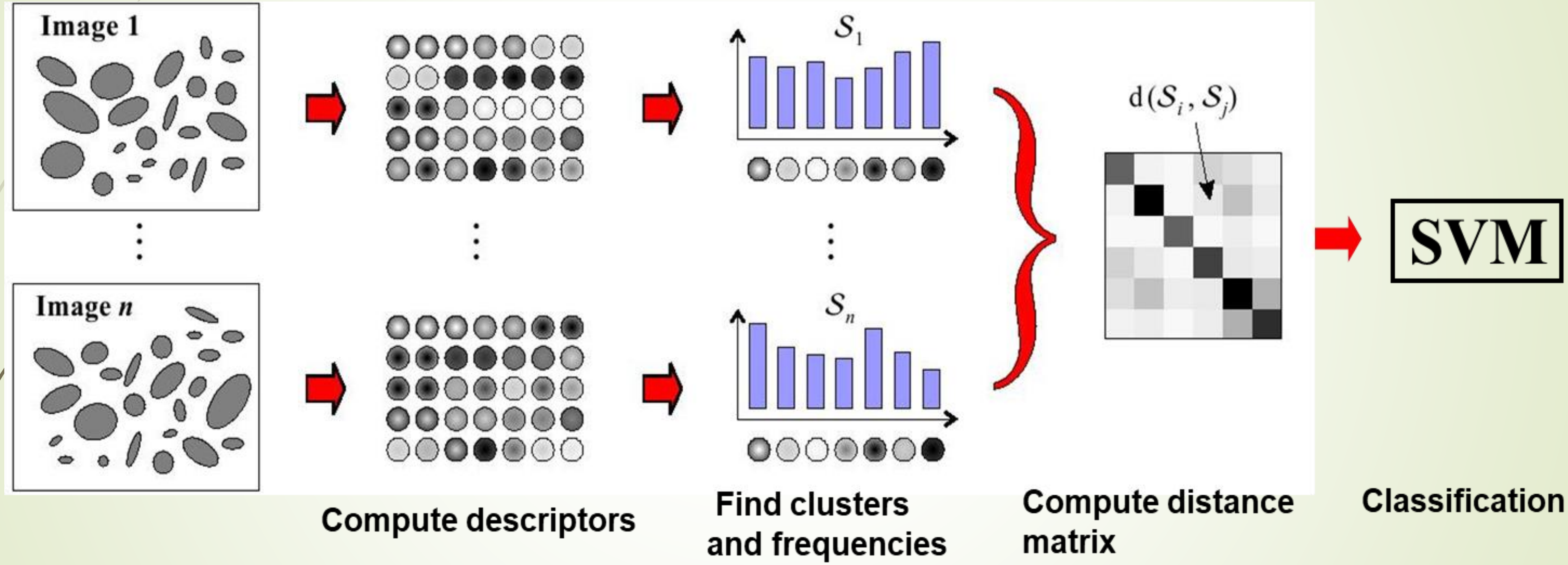


(Csurka, Dance, Perronnin et al. 2006)

bikes books building cars people phones trees

Category Recognition

Bag –of-Feature for Image Classification



feature detectors (Harris–Laplace and Laplacian, etc.)
descriptors (SIFT, RIFT, etc.)
classifier (naive Bayesian and support vector machines, etc.)

Scene Understanding

- *Scene understanding* identify the target of objects and also the distribution of targets in a scene.
- It requires reasoning about both *what we can see* and *what is occluded*.
- It is a challenging and most important problem in computer vision.
- Scene understanding have a significant impact in computer vision to perceive, analyze, and interpret the visual scenes which leads to new research areas.

Scene Understanding

Less-Successful Recognition Results

Simultaneous recognition and segmentation using TextonBoost

Scene Understanding

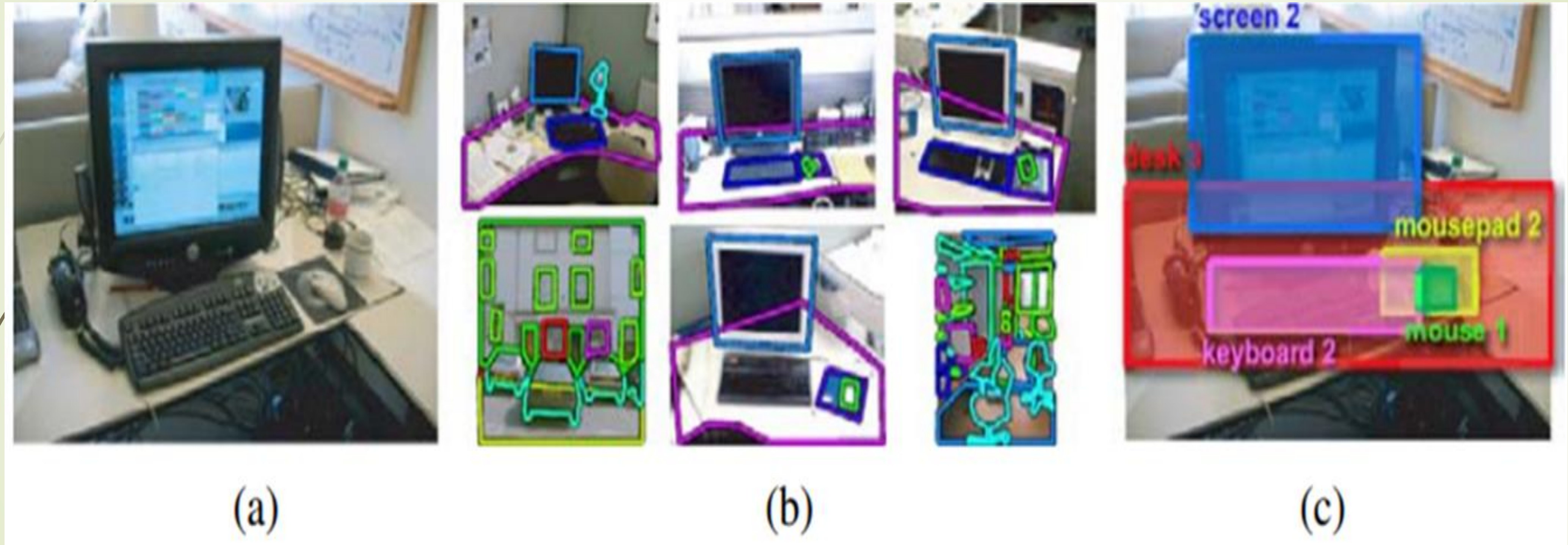
Learning and Large Image Collections

- Early learning-based algorithms, used relatively few (in the hundreds) labeled examples to train recognition algorithm parameter.
- Today, some recognition algorithms use databases such as LabelMe which contain tens of thousands of labeled examples.
- The existence of such large databases opens up the possibility of matching directly against the training images.
- From which a consensus labeling for the unknown objects in the scene can be inferred

Scene Understanding

Recognition by Scene Alignment

- (Russell, Torralba, Liu et al. 2007)

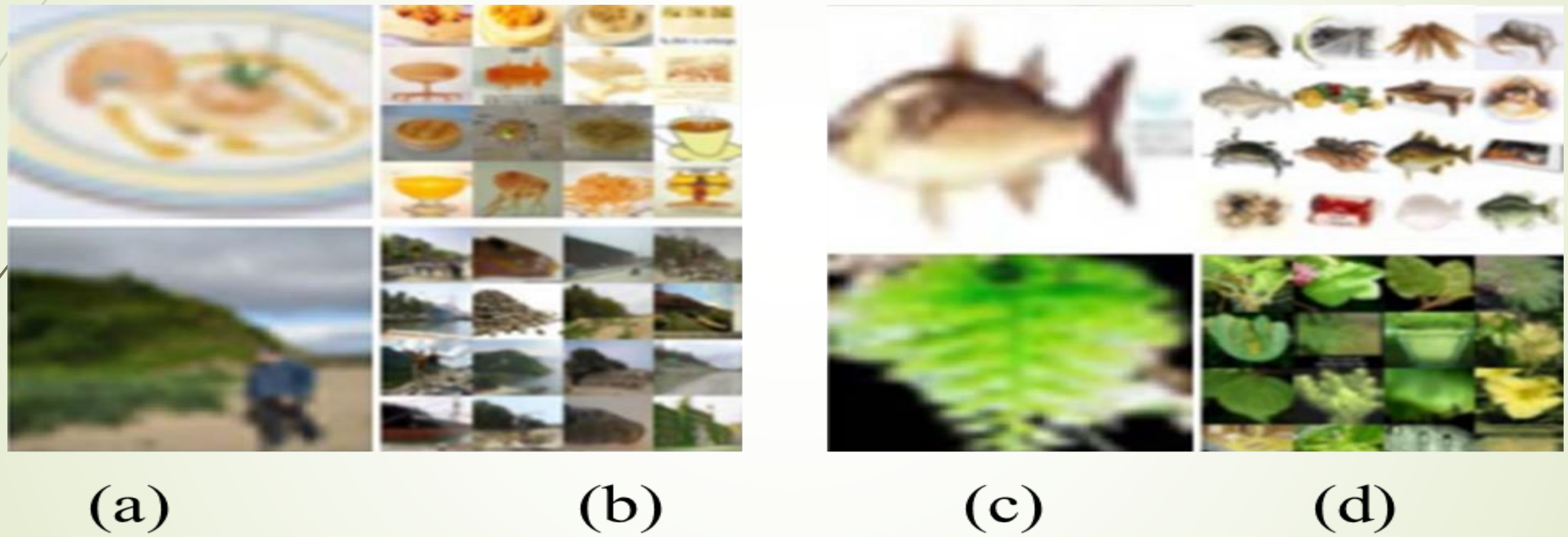


(a) input image; (b) matched images with similar scene configurations; (c) final labeling of the input image.

Scene Understanding

Recognition by Tiny Images

- (Torralba, Freeman, and Fergus 2008)

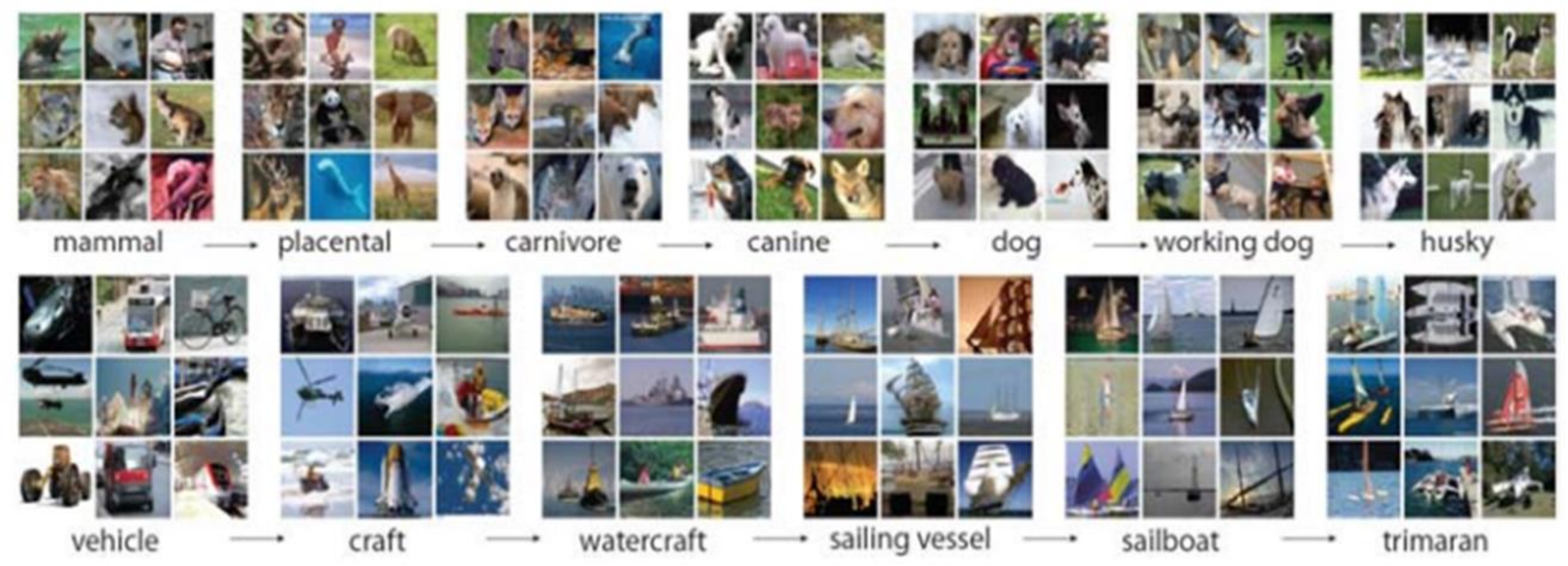


(a) and (c) show sample input images and columns
(b) and (d) show the corresponding 16 nearest neighbors in the database of 80 million tiny images

Scene Understanding

ImageNet

- (Deng, Dong, Socher et al. 2009)

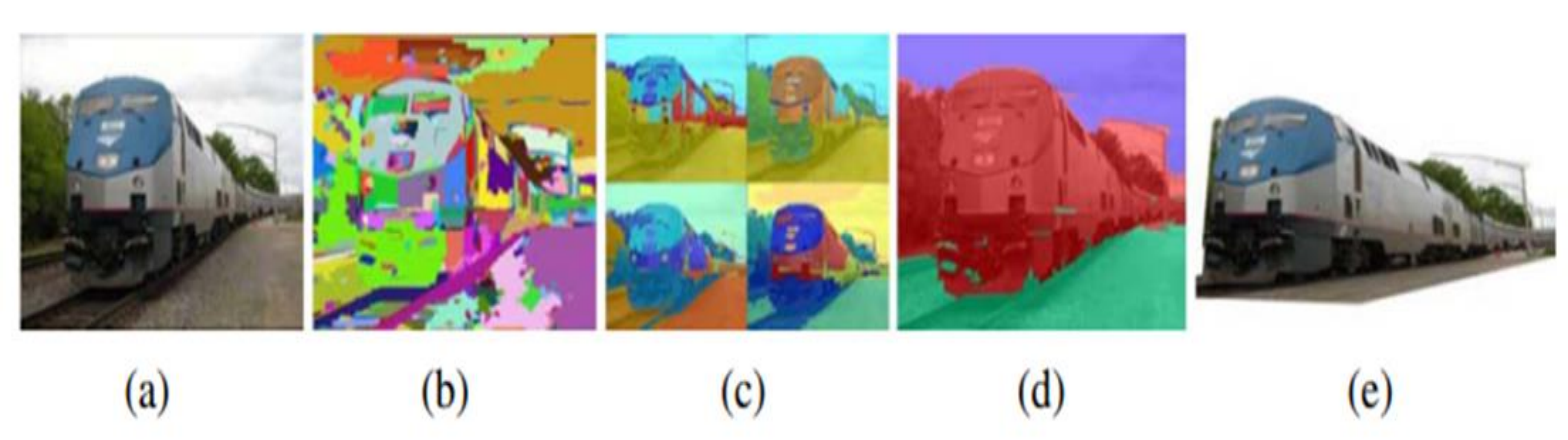


This database contains over 500 carefully vetted images for each of 14,841 nouns from the WordNet hierarchy.

Scene Understanding

Application: Intellignet photo editing

- (Hoiem, Efros, and Hebert 2005)



(a) input image; (b) superpixels are grouped into (c) multiple regions;

(d) labelings indicating ground (green), vertical (red), and sky (blue);

(e) novel view of resulting piecewise-planar 3D model.